AquaNexis — Round 2 Engineering Progress Report

AtomQUEST 2026 | Track 2: Water Purifier | Institution: UPES Dehradun | Solo Prototype | Date: 4 November 2026

1. Concise Abstract & Exploration Scope

Abstract: AquaNexis is a software-only Predictive Water Intelligence System built for AtomQUEST 2026 Track 2 (Water Purifier). Conventional purifiers rely on rigid calendar timers or single-point TDS cutoffs that cause either premature filter disposal or undetected water quality degradation. AquaNexis implements multi-parameter input water quality sensing beyond TDS and prognostics of filter failure. Operating in verified DEMO_MODE, it computes a 7-component Filter Stress Index (FSI), 5-component Filter Health Index (FHI), deterministic Remaining Useful Life (RUL) with explicit fallback labeling (MODEL_MISSING), and failure risk classification with counterfactual explanations. No drinking-water certification is claimed; physical actuator shutdown is deferred to Round 3.

Dimension	Technical Specification & Exploration Scope
Track & Focus	Track 2 — Water Purifier | Institution: UPES Dehradun | Prototype Stage: Software Intelligence System
Primary Exploration	Prognostics of filter failure using data analytics (FSI, FHI, RUL, Failure Risk, Explainability Engine)
Secondary Exploration	Input water quality sensing beyond TDS (TDS, EC, pH, Turbidity, Temp, Flow, Transmembrane Delta-P)
Provenance Standard	MEASURED (hardware) | SIMULATED (simulator/demo) | CALCULATED (derived) | MODEL_PREDICTED (ML/prognostics)

2. Problem, Solution & Completed Modules

Core Pipeline: Sense (13 sensor channels) → Understand (QualityValidator & AquaScore) → Predict (FSI, FHI, RUL) → Alert (State Machine) → Assist (Maintenance Engine).

Module	Status	Verified Technical Capabilities & Grounded Reality
Telemetry Ingestion & Security	Complete	FastAPI ingestion, monotonic sequence replay defense, clock-skew (+/-300s) checks, JWT claims verification
Water Quality (Beyond TDS)	Complete	AquaScore (0-100) aggregating TDS, EC, pH, Turbidity, Temp; 1h trend slopes & anomaly z-scores
Filter Stress Index (FSI)	Complete	7-component weighted stress index; dynamic denominator excludes missing sensors without inventing values
Filter Health Index (FHI)	Complete	5-component health index (0-100) mapped to maintenance bands: NORMAL, MONITOR, REPLACE_RECOMMENDED
RUL Prognostics & ML Pipeline	Complete	Controlled dataset (controlled-membrane-deg-v1.0), multivariate model (MAE 4.85d), fallback MODEL_MISSING
Explainability & Counterfactuals	Complete	Top 5 contributing degradation features (delta-P, flow deviation, volume, age, turbidity) + replacement recovery
Alert State Machine	Complete	Formal lifecycle: DETECTED → OPEN → ACKNOWLEDGED / SNOOZED / RESOLVED / AUTO_CLEARED with evidence drawer
Maintenance Workflows	Complete	11-category catalog, replacement guide, sensor calibration tracker, and post-service verification runs
Web UI & Live WebSocket	Complete	React 18 + Vite + TypeScript dashboard (localhost:3000), realtime WebSocket, scenario switcher

3. Sourcing Risks, Scope Reset & Budget (Stage 1 Estimate vs Round 2 Actual)

Item / Sourcing Risk	Stage 1 Estimate	Round 2 Actual	Engineering Decision & Honest Disclosure
ESP32 + 13 Sensors + PCB	Rs. 5,000 - 15,000	Rs. 0 (Not purchased)	Hardware deferred to eliminate procurement risk; firmware/esp32/ preserved as reference
Hydraulic Bench & Solenoids	Bench rig planned	Rs. 0 (Software-only)	Replaced by deterministic 8-profile Python simulator (backend/simulator.py)
TimescaleDB Extension	Local PG16 + Timescale	Rs. 0 (DEMO_MODE)	Windows lack of timescaledb DLL resolved by isolated DEMO_MODE; CI validates PG16 container

4. Performance Targets & Grounded Verification Status

Metric / Capability	Target Specification	Observed Verification Result & Explicit Disclosures
Contextual Signals Beyond TDS	>=4 parameters beyond TDS	Verified: 13 channels (TDS, EC, pH, Turbidity, Temp, Flow, Inlet/Outlet Pressure, Tank, Leak, Current, Power, UV)
RUL Prognostic Accuracy	RUL MAE <= 7 days	Verified on controlled dataset: MAE 4.85 days (R2=0.9962); fallback labeled MODEL_MISSING until empirical field trials
Dashboard Telemetry Freshness	Update cadence <= 5 seconds	Verified: WebSocket broadcast latency <100ms; UI stale badge activates when age exceeds 30 seconds
Actuator Safety & Fail-safe	Automatic physical isolation	Disclosed: Software alert & audit log only. Leak detection creates a critical safety event and records/recommends an isolation action. Physical actuator shutdown is not claimed for Round 2; hardware-in-loop validation is planned for Round 3.
Sensor Fault Resilience	Graceful quality degradation	Verified: SENSOR_FAULT flags Quality=BAD, drops missing channels from denominator, withholds RUL (Confidence=LOW)

5. Representative Scenario Demonstration Results (Reproduced in DEMO_MODE)

Scenario	Telemetry & Quality State	Intelligence Response (AquaScore / FSI / FHI / RUL / Risk)	Alert & Safety Action Taken
NORMAL	Quality: GOOD, SIMULATED dP=30 kPa, Flow=2.0 L/m	AquaScore=100 (EXCELLENT), FSI=20.0, FHI=86% (NORMAL), RUL=154.8d (MODEL_MISSING), Risk=LOW	0 active alerts; routine background telemetry
FILTER_STRESS	Quality: GOOD, SIMULATED dP=176.9 kPa, Flow=0.96 L/m	AquaScore=93, FSI=67.5 (HIGH), FHI=37% (POOR), RUL=66.6d (MODEL_MISSING), Risk=MEDIUM, Band=REPLACE_RECOMMENDED	Alert: filter_stress_high (CRITICAL); Stage 1 Sediment cartridge replacement recommended
LEAK	Quality: GOOD, SIMULATED leak_detected=True	AquaScore=100, FSI=20.0, FHI=86%, Risk=LOW, Safety Status=CRITICAL_EVENT	Alert: leak_detected (CRITICAL); isolation action logged/recommended; NO physical actuation
SENSOR_FAULT	Quality: BAD, SIMULATED Turbidity out-of-range, missing P_out	AquaScore=90, FSI dynamic denominator active, RUL withheld (N/A, LOW confidence)	Alert: sensor_fault_detected (WARNING); corrupt data prevented from driving false prediction

6. Automated Verification, Safety Disclosures & Round 3 Roadmap

Automated Test Evidence: Backend unit tests: **154/154 PASS (11 skipped DB-dependent)** in 1.87s across configuration, security, canonical telemetry, ML RUL, hydraulic intelligence, alerts, and maintenance. Frontend checks: **npm run lint PASS** (0 errors, 0 warnings), **npm run format:check PASS**, **npm run typecheck PASS** (0 errors), **npm test PASS (14/14)**, **npm run build PASS** (production bundle produced).

Honest Engineering Disclosures: (1) No drinking-water certification (e.g. BIS/NSF/WQA) is claimed; AquaNexis is an operational predictive intelligence system; (2) All demonstration telemetry is SIMULATED via backend/simulator.py; (3) Actuator isolation is recorded as a software recommendation; physical solenoid valve actuation is not claimed for Round 2; (4) Local demo runs without external database via DEMO_MODE; production TimescaleDB hypertables are verified in CI.

Round 3 Roadmap: (1) Long-term empirical run-to-failure membrane aging on a physical test bench to replace synthetic dataset training; (2) Hardware-in-the-loop (HIL) safety certification for physical Normally Closed solenoid valve actuators; (3) Cloud-hosted TimescaleDB deployment for multi-device fleet telemetry and OTA calibration updates.